

TTT-01 · BIOLOGY + ENERGY

MudWatt Bioelectric League

Can your team make mud generate electricity and defend your design with data?

PREDICT > BUILD or SOLVE > TEST > SCORE > REDESIGN

WIN: Highest combined score from peak reading, daily data quality, controlled-variable design, redesign notes, and final explanation.

TTT-02 · FIELD SCIENCE

Forest Sensor Sprint

Find the smartest place to install a future campus tree sensor.

PREDICT > BUILD or SOLVE > TEST > SCORE > REDESIGN

WIN: Best evidence-based sensor recommendation with accurate measurements and efficient route planning.

TTT-03 · BIOMIMICRY

Seed Dispersal Derby

Build a seed that escapes the parent tree without engines or throwing.

PREDICT > BUILD or SOLVE > TEST > SCORE > REDESIGN

WIN: Best combined distance, hang time, target landing, biomimicry explanation, and redesign improvement.

TTT-04 · BIOMIMICRY

Xylem Pipeline Relay

Move water uphill like a tree, faster and cleaner than rival teams.

PREDICT > BUILD or SOLVE > TEST > SCORE > REDESIGN

WIN: Most water delivered with speed, low spill, and strong explanation.

TTT-05 · ECOLOGY

Greenhouse Climate Controller

Run the greenhouse controls for a plant that cannot complain, only wilt.

PREDICT > BUILD or SOLVE > TEST > SCORE > REDESIGN

WIN: Best plant-zone-control match supported by tour evidence.

TTT-06 · BIOMIMICRY

Pinecone Weather Machine

Build a weather sensor with no battery and no code.

PREDICT > BUILD or SOLVE > TEST > SCORE > REDESIGN

WIN: Most accurate, readable, and fast response after spray-and-dry trials.

TTT-07 · ECOLOGY

Pollinator Network Draft and Build

Draft a pollinator habitat that works all season, not just on the prettiest day.

PREDICT > BUILD or SOLVE > TEST > SCORE > REDESIGN

WIN: Highest pollinator support across spring, summer, and fall using native and clumping logic.

TTT-08 · FIELD SCIENCE

Arboretum Eco-Quest

Read the living collection and solve the checkpoints before time runs out.

PREDICT > BUILD or SOLVE > TEST > SCORE > REDESIGN

WIN: Most accurate checkpoint solutions with safe field conduct.

TTT-09 · ECOLOGY

Minecraft Tree World Resilience Cup

Design a landscape that survives stress because of how it is built, not luck.

PREDICT > BUILD or SOLVE > TEST > SCORE > REDESIGN

WIN: Best climate-resilient design with accurate nature-based strategies.

TTT-10 · DATA + CLIMATE

Tree Ring Climate Detective

Read the rings to reconstruct the climate events a tree lived through.

PREDICT > BUILD or SOLVE > TEST > SCORE > REDESIGN

WIN: Most accurate climate-event inferences with claim-evidence-reasoning.

TTT-11 · MATERIALS

Lotus Leaf Surface Sprint

Build a surface that sheds water and dirt the way a lotus leaf does.

PREDICT > BUILD or SOLVE > TEST > SCORE > REDESIGN

WIN: Best clean runoff and least residue after one redesign.

TTT-12 · FIELD SCIENCE

Leaf Stomata Microscope Detective

Count the breathing pores and rank leaves by how they manage water.

PREDICT > BUILD or SOLVE > TEST > SCORE > REDESIGN

WIN: Most accurate detective ranking with clean data and strong evidence.

TTB-01 · FIELD SCIENCE

Root Grip Erosion Rescue

Can roots hold a slope together during a storm?

PREDICT > BUILD or SOLVE > TEST > SCORE > REDESIGN

WIN: Most soil retained after standard pour.

TTB-02 · FIELD SCIENCE

Tree Height Triangulation Shootout

Measure a tree without climbing it.

PREDICT > BUILD or SOLVE > TEST > SCORE > REDESIGN

WIN: Closest to instructor reference height.

TTB-03 · DATA + CLIMATE

Urban Heat Shade Dash

Find the coolest route through a hot campus.

PREDICT > BUILD or SOLVE > TEST > SCORE > REDESIGN

WIN: Best data-backed cool route.

TTB-04 · FIELD SCIENCE

Photosynthesis Float-Off Playoffs

Make leaf disks float by producing oxygen.

PREDICT > BUILD or SOLVE > TEST > SCORE > REDESIGN

WIN: Fastest half-float time with controlled variables.